

Traffic Signal Control via Detection-Driven Vehicle Counting

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OBJECTIVE

Fixed-cycle traffic lights hold a green light for an empty lane while a queue builds beside it. This project replaces the fixed cycle with signal timing driven by how many vehicles are actually waiting at the intersection.

APPROACH

A YOLOv11 detector (YOLO11m, 20.1M parameters) detects and counts vehicles inside a bounded region of the frame. Entry and exit lines increment and decrement a live queue count. That count maps to a green-light duration through a fixed rule set — for example, a queue of 8 vehicles receives 20 seconds of green, a queue of 20 receives 30 — with no learned control policy behind it. A four-way intersection modelled in SUMO, with two incoming and two outgoing lanes per direction, served as the test bed across light, moderate and heavy traffic scenarios. A Streamlit application ties the pieces together with a dataset viewer, a live prediction view, and a simulation runner.

DATASET & TRAINING

Roboflow-sourced imagery and video frames across varied times of day and weather, pre-labelled in YOLO format. Split 70% training / 15% validation / 15% test, 640×640 images, batch size 32, 100 epochs.

RESULTS — PROPOSED MODEL VS. BASELINE

Metric	Proposed	Baseline
Precision	0.92	0.81
Recall	0.90	0.45
F1 (operating threshold)	0.91	0.46
mAP@0.5	0.96	0.45
mAP@0.5–0.95	0.76	0.31

The largest gain is in recall: 0.45 to 0.90. The baseline was missing roughly half of all vehicles, and in a system that allocates green time by vehicle count, a missed detection becomes wrong timing — the reason recall was tracked more closely here than headline accuracy.

Note on F1: Ultralytics YOLO also reports an F1–confidence curve. Its peak value was 0.94 at an optimised confidence threshold; the value shown above (0.91) is at the same operating threshold as the precision and recall figures, and is the one directly comparable to them.

CONTRIBUTION

Team project. Worked across all four workstreams: detector training and evaluation, the queue-manager and entry/exit counting logic, the SUMO network and traffic scenarios, and the Streamlit interface. Simulation only — never deployed to a real intersection.

BUILT WITH

YOLOv11 · SUMO · Streamlit · Python